

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ SOLUTIONS

Chapter 2: Polynomials

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Zeroes and coefficients of a quadratic polynomial

1. The zeroes α and β of the polynomial $3x^2 + 6x + k$ satisfy $\alpha + \beta + \alpha\beta = -\frac{2}{3}$. Find k .

CBSE 2025, 1 mark

SOLUTION

$$\alpha + \beta = -\frac{6}{3} = -2 \text{ and } \alpha\beta = \frac{k}{3}$$

$$-2 + \frac{k}{3} = -\frac{2}{3}, \text{ so } \frac{k}{3} = \frac{4}{3}$$

Answer: $k = 4$

2. Find the zeroes of the polynomial $x^2 + \frac{4}{3}x - \frac{4}{3}$. CBSE 2025, 2 marks

SOLUTION

$$x^2 + \frac{4}{3}x - \frac{4}{3} = \frac{1}{3}(3x^2 + 4x - 4)$$

$$3x^2 + 4x - 4 = 3x^2 + 6x - 2x - 4 = 3x(x + 2) - 2(x + 2) = (3x - 2)(x + 2)$$

The polynomial is zero when $3x - 2 = 0$ or $x + 2 = 0$.

Check: sum $= \frac{2}{3} - 2 = -\frac{4}{3}$ and product $= \frac{2}{3} \times (-2) = -\frac{4}{3}$, which match the coefficients.

Answer: $\frac{2}{3}$ and -2

3. The zeroes of $p(x) = 2x^2 - k\sqrt{2}x + 1$ add up to $\sqrt{2}$. Find k . (a) $\sqrt{2}$ (b) 2 (c) $2\sqrt{2}$ (d) $\frac{1}{2}$

CBSE 2024, 1 mark

SOLUTION

$$\text{Sum of zeroes} = -\frac{b}{a} = \frac{k\sqrt{2}}{2}$$

$$\frac{k\sqrt{2}}{2} = \sqrt{2} \text{ gives } k = 2$$

Answer: (b)

4. Each zero of $x^2 + px + q$ is double a zero of $4x^2 - 5x - 6$. Find p . (a) $-\frac{5}{2}$ (b) $\frac{5}{2}$ (c) -5 (d) 10

CBSE 2024, 1 mark

SOLUTION

Let the zeroes of $4x^2 - 5x - 6$ be α and β . Then $\alpha + \beta = \frac{5}{4}$.

The zeroes of $x^2 + px + q$ are 2α and 2β , with sum $2(\alpha + \beta) = \frac{5}{2}$.

Sum of zeroes $= -p$, so $p = -\frac{5}{2}$

Answer: (a)

2 Zeroes from the graph

5. The graphs of two quadratic polynomials are drawn on the same axes. One parabola opens upwards and the other opens downwards, and both cut the x -axis at the same two points. How many distinct zeroes do the two polynomials have between them? CBSE 2025, 1 mark

SOLUTION

The zeroes of a polynomial are the x -coordinates of the points where its graph meets the x -axis.

Both graphs meet the x -axis at the same two points, so they share the same two zeroes.

Answer: 2

6. Assertion (A): a polynomial whose graph touches the x -axis at exactly one point cannot be quadratic. Reason (R): a polynomial of degree n , with $n > 1$, has at most n zeroes. Choose: (a) A and R are both true and R explains A. (b) A and R are both true but R does not explain A. (c) A is true and R is false. (d) A is false and R is true. CBSE 2024, 1 mark

SOLUTION

A is false: the graph of $(x - 1)^2 = x^2 - 2x + 1$ touches the x -axis only at $x = 1$, and it is a quadratic polynomial.

R is true: a polynomial of degree n has at most n zeroes.

Answer: (d)

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